

Sets

$$A - B = A \cap B'$$

To prove that : $A \cap (B - C) = (A \cap B) - (A \cap C)$

$$\Rightarrow x \in A \cap (B - C)$$

$$\Rightarrow x \in A \text{ \& } x \in B - C$$

$$\Rightarrow x \in A \text{ \& } x \in B \text{ \& } x \notin C$$

$$\Rightarrow x \in A \text{ \& } x \in B \text{ \& } x \in A \text{ \& } x \notin C$$

$$\Rightarrow x \in (A \cap B) \text{ \& } x \notin (A \cap C)$$

$$\Rightarrow x \in (A \cap B) - (A \cap C)$$

To prove

$$A = B$$

$$A \subseteq B \text{ \& }$$

$$B \subseteq A$$

As we gone from left to right, similarly we can go from right to left. (or up to down & vice versa)

Cartesian product of sets

$$A \times B = \{ (a, b) : a \in A, b \in B \}$$

$$1) A \times (B \cap C) = (A \times B) \cap (A \times C)$$

Let $x, y \in A \times (B \cap C)$

$$\Rightarrow x \in A \text{ \& } y \in B \cap C$$

$$\Rightarrow x \in A \text{ \& } y \in B \text{ \& } y \in C$$

$$\Rightarrow x \in A \text{ \& } y \in B \text{ \& } x \in A \text{ \& } y \in C$$

$$\Rightarrow x \in (A \cap A) \text{ \& } (x, y) \in (A \times B) \cap (A \times C)$$

$$(x, y) \in (A \times B)$$

$$\Rightarrow (x, y) \in (A \times B) \cap (A \times C)$$

gmp.

$$2) \quad A \times (B \cup C) = (A \times B) \cup (A \times C)$$

$$\Rightarrow \text{Let } x, y \in A \times (B \cup C)$$



$$\Rightarrow x \in A \text{ \& } y \in B \cup C$$

$$\Rightarrow x \in A \text{ \& } y \in B \text{ or } y \in C$$

$$\Rightarrow x \in A \text{ \& } y \in B \text{ or } x \in A \text{ \& } y \in C$$

$$\Rightarrow (x, y) \in A \times B \text{ or } (x, y) \in A \times C$$

$$\Rightarrow (x, y) \in (A \times B) \cup (A \times C)$$

$$3) \quad A \times (B - C) = (A \times B) - (A \times C)$$

$$\Rightarrow \text{Let } x, y \in A \times (B - C)$$

$$\Rightarrow x \in A \text{ \& } y \in B - C$$

$$\Rightarrow x \in A \text{ \& } y \in B \text{ \& } y \notin C$$

$$\Rightarrow x \in A \text{ \& } y \in B \text{ \& } x \in A \text{ \& } y \notin C$$

$$\Rightarrow (x, y) \in (A \times B) \text{ \& } (x, y) \notin A \times C$$

$$\Rightarrow (x, y) \in (A \times B) - (A \times C).$$

Functions:

Find inverse of $f(x) = 2x - 5$, if possible where
 $f: \mathbb{R} \rightarrow \mathbb{R}$

$$\text{Let } f(x_1) = f(x_2)$$

$$2x_1 - 5 = 2x_2 - 5$$

$$x_1 = x_2$$

f is 1-1 (injective)

$$\text{Let } y \in \mathbb{R}, \quad y = 2x - 5$$

$$x = \frac{y+5}{2} \in \mathbb{R}$$

$\Rightarrow \forall y \in \mathbb{R}$ (range) $\exists$ a preimage $x \in \mathbb{R}$ (domain)

$\Rightarrow f$ is onto (surjective)

Q. $f: \mathbb{R} - \{-1\} \rightarrow \mathbb{R} - \{1\}$ defined by $f(x) = \frac{x}{x+1}$

$$\text{Let } f(x_1) = f(x_2)$$

$$\frac{x_1}{x_1+1} = \frac{x_2}{x_2+1}$$

$$\cancel{\frac{1}{x_1+1}} = \cancel{\frac{x_2}{x_2+1}}$$

$$x_1+1 = x_2+1$$

$$x_1 = x_2$$

$$\text{Let } y \in \mathbb{R} - \{1\}$$

$$y = \frac{x}{x+1} \Rightarrow x = \frac{y}{1-y} \in \mathbb{R} - \{-1\}$$

$\forall y \in \mathbb{R} - \{-1\} \exists$ unique primary
 $x \in \mathbb{R} - \{-1\}$

$\therefore f$ is onto

$$f^{-1}(x) = \frac{x}{1-x}$$

Q $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^2$

$$f^{-1}(3) = x$$

$$f(x) = 3$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$

$$f^{-1}(3) = \{-\sqrt{3}, \sqrt{3}\}$$

$$f^{-1}(10) = x$$

$$f(x) = 10$$

$$x^2 + 1 = 10$$

$$x^2 = 9$$

$$x = \pm 3$$

$$f^{-1}(10) = \{-3, 3\}$$

$$f(x) = x^2 - 4$$

$$f^{-1}(-5) =$$

$$-5 = x^2 - 4$$

$$-1 = x^2 \Rightarrow x = \pm i$$

$\hookrightarrow$ Not for real

$$x^2 - 4 = -2$$

$$x^2 = 2 \Rightarrow x = \pm\sqrt{2}$$

Relations

$$R \subseteq A \times B$$

$$R = \{ (a, b) : a \in A, b \in B, x R y \}$$

x is related to y

$$F = \{ (a, f(a)) : x \in a, y \in B \}$$

* If x is related to y by a relation R then we write $x R y$ or $(x, y) \in R$.

* If $\#(A) = m$, $\#(B) = n$ then there can be at most 2^{mn} relations from A to B .

Ex- Let $A = \{2, 4, 6, 8\}$, $B = \{1, 3, 5, 7\}$, $R \subseteq A \times B$

1) $x R y$ iff $x < y$ $R = \{ (2, 3), (2, 5), (2, 7), (4, 5), (4, 7), (6, 7) \}$

2) $x R y$ iff $x = y + 1$ $R = \{ (2, 1), (4, 1), (6, 5), (8, 7) \}$

3) $x R y$ iff $x + y$ is divisible by 3
 $R = \{ (2, 1), (2, 7), (4, 5), (6, 3), (8, 1), (8, 7) \}$

Similarly, many other relations can be defined from A to B .

* These are called Binary Relations.

* If x is not related to y via relation R then we write $x \not R y$ or $(x, y) \notin R$.

Types of relations

1. Reflexive Relation: Let R be a relation on set A .
Then R is reflexive if $xRx \forall x \in A$
or $(x, x) \in R$

eg. Let $A = \{1, 2, 3\}$ & $R = \{(1, 2), (1, 1), (2, 2), (3, 2)\}$ then R is not reflexive as $(3, 3) \notin R$.

2. Symmetric Relation: A relation R on set A is said to be symmetric if aRb then bRa or if $(a, b) \in R$ then $(b, a) \in R$; $a, b \in A$

eg. $A = \{1, 2, 3\}$ $R = \{(1, 2), (1, 1), (3, 3), (2, 1)\}$ is a symmetric relation.

3. Anti-Symmetric relation: A relation R on set A is said to be antisymmetric if aRb & bRa , then $a=b$ or, if $(a, b) \in R$ & $(b, a) \in R$, then $a=b$.

4. Transitive relation: Let R be relation on set A .
Then R is called transitive iff
 aRb and bRc then aRc , i.e., $(a, b) \in R$,
 $(b, c) \in R \Rightarrow (a, c) \in R$.

Ex- If $A = \{1, 2, 3\}$ & $R = \{(1, 2), (2, 3), (1, 3)\}$ then R is transitive.
 $\therefore (1, 2), (2, 3) \in R$ & $(1, 3) \in R$

Q $R = \{(1,2), (2,2), (3,4), (4,3), (2,3), (3,2), (3,3)\}$

- (a) Not reflexive as $(1,1) \notin R$ & $(4,4) \notin R$
- (b) Not symmetric as $(1,2) \in R$ & $(2,1) \notin R$
- (c) Not transitive as $(4,3) \in R$ & $(3,4) \in R$ but $(4,4) \notin R$
- (d) Not anti-symmetric as $(3,4) \in R$ & $(4,3) \in R$

Q $R = \{(1,2), (2,1), (1,1), (2,2), (3,4)\}$

- (a) Not reflexive as $(3,3) \notin R$ & $(4,4) \notin R$
- (b) Not symmetric as $(3,4) \in R$ but $(4,3) \notin R$
- (c) Transitive
- (d) Not anti-symmetric as $(1,2) \in R$ & $(2,1) \in R$

Q Give an example of relation which is

- (a) reflexive but not symmetric & transitive
- (b) symmetric but not reflexive and transitive
- (c) transitive and reflexive but not symmetric

Solⁿ (a) $\{(1,1), (2,2), (3,3), (1,2), (2,3)\}$

(b) $\{(1,2), (2,1), (3,3)\}$

(c) $\{(1,2), (2,1), (1,1), (2,2), (3,3), (1,3), (2,3)\}$

Q. Let R be a relation on set of all integers. Determine whether R is reflexive, symmetric, antisymmetric or transitive when $(x,y) \in R$ iff

- (a) $xy > 1$
- (b) $|y-x| + 2$ is a prime

(a) xRy iff $xy > 1$

(i) Not reflexive as $(1,1) \notin R$

(ii) Symmetric $\rightarrow$ Yes

- iii) Transitive $\rightarrow$ Yes
 iv) Antisymmetric $\rightarrow$ No

(b) $(x, y) \in R$ iff $|y - x| \pm 2$ is prime

- 1) Reflexive $\rightarrow$ Yes
 2) Symmetric $\rightarrow$ Yes
 3) Transitive $\rightarrow$ No
 4) Antisymmetric $\rightarrow$ No

Equivalence relation: A relation on set A is said to be an equivalence

relation if it is

- (1) Reflexive (2) Symmetric (3) Transitive

Ex- Let $\mathbb{Z}_0$ denote set of all non-zero integers and R be a relation defined by $x^y = y^x$ where $x, y \in \mathbb{Z}_0$. Is R an equivalence relation?
 $R = \{(x, y) : x^y = y^x, x, y \in \mathbb{Z}_0\}$

Solⁿ: 1) Reflexive

Let $x \in \mathbb{Z}_0$. Then $x^x = x^x \Rightarrow (x, x) \in R$
 $\forall x \in \mathbb{Z}_0$

$\therefore$ Reflexive

2) Symmetric

Let $(x, y) \in R \Rightarrow x^y = y^x$
 $\Rightarrow y^x = x^y \Rightarrow (y, x) \in R$
 Hence, symmetric

(3) Transitive

$$\begin{aligned} \text{Let } (x, y), (y, z) \in R &\Rightarrow xy = yx \text{ \& } yz = zy \\ &= (xy)z = (yx)z \text{ and } (yz)x = (zy)x \\ &= (xz)y = (zy)x \quad \text{--- (1)} \\ &= (yz)x = (zx)y \quad \text{--- (2)} \end{aligned}$$

$$\text{From (1) \& (2), } \therefore (xz)y = (zx)y \Rightarrow xz = zx \\ \Rightarrow (x, z) \in R$$

Hence, transitive

Hence, it is an equivalence relation.

Ex- Let $\mathbb{Z}$ be a set of integers and $X = \mathbb{Z} \times \mathbb{Z}$.
Define relation $\sim$ on X as $(a, b) \sim (c, d)$ if $ad = bc$.
Prove that $\sim$ is an equivalence relation.

Solⁿ: 1) Let $(a, b) \in X$. Now, $ab = ba$ ($\because a, b \in \mathbb{Z}$)
 $\rightarrow (a, b) \sim (a, b)$
 $\sim$ is reflexive.

2) Let $(a, b), (c, d) \in X$

$$\begin{aligned} \text{Let } (a, b) \sim (c, d) &\Rightarrow ad = bc \\ &\Rightarrow bc = ad \\ &\Rightarrow (c, d) \sim (a, b) \Rightarrow \sim \text{ is} \\ &\text{symmetric} \end{aligned}$$

3) Let $(a, b), (c, d), (e, f) \in X$
 $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$
 $\Rightarrow \frac{a}{b} = \frac{c}{d} \quad \Rightarrow \frac{cf}{d} = \frac{e}{f}$

$$\Rightarrow \frac{a}{b} = \frac{e}{f} \Rightarrow af = be \Rightarrow (a, b) \sim (e, f)$$

$\rightarrow \sim$ is transitive.

Hence, $\sim$ is an equivalence relation.

Ex- Let R be a relation on set of integers $\mathbb{Z}$ defined by $a \equiv b \pmod{3}$. Show that it is an equivalence relation.

$$\Rightarrow a \equiv b \pmod{3} \Rightarrow a - b \text{ is a multiple of } 3 \\ \text{i.e., } a - b = 3k, \quad k \in \mathbb{Z}$$

$$(i) \text{ Reflexive: } x \equiv x \pmod{3} \text{ as } x - x = 0(3) \\ 0 \in \mathbb{Z}$$

$$\Rightarrow (x, x) \in R \quad \forall x \in \mathbb{Z}. \text{ Hence, reflexive.}$$

$$(ii) \text{ Symmetric: } (x, y) \in R \Rightarrow x - y = 3k, \quad k \in \mathbb{Z} \\ y - x = 3(-k), \quad -k \in \mathbb{Z}$$

$$\Rightarrow (y, x) \in R. \text{ Hence, symmetric.}$$

$$(iii) \text{ Transitive: } (x, y), (y, z) \in R \Rightarrow x - y = 3k, \\ y - z = 3l, \\ k, l \in \mathbb{Z}$$

$$\therefore x - z = 3k + 3l = 3(k + l)$$

$$\& \quad k + l \in \mathbb{Z}$$

$$\Rightarrow (x, z) \in R \text{ Hence transitive.}$$

$$\Rightarrow a \equiv b \pmod{3} \text{ is an equivalence relation of } \mathbb{Z}$$

Equivalence Class: Let R be an equivalence relation in set A . Then set of all elements of A that are related to x is called equivalence class of x under R .

$$[x] = \{y: y \in A \text{ and } (x, y) \in R\}$$

Note: $A/R = \{[x]: x \in A\} \rightarrow$ Set of all equivalence classes of elements of A under equivalence relation R is called quotient set of A by R .

Ex Find distinct equivalence classes of R where $R = \{(x, y): x, y \in \mathbb{Z}, xRy \Leftrightarrow x \equiv y \pmod{3}\}$

$\Rightarrow$ We have already proved that R is an equivalence relation.

$$\begin{aligned} \text{Let } [a] &= \{x \in \mathbb{Z}: xRa\} \\ &= \{x \in \mathbb{Z}: x-a \text{ is divisible by } 3\} \\ &= \{x \in \mathbb{Z}: x-a=3k, \text{ for some integer } k\} \\ &= \{x \in \mathbb{Z}: x=a+3k \text{ for } k \in \mathbb{Z}\} \end{aligned}$$

Here, $[a]$ is equivalence class generated by $a \in \mathbb{Z}$

$$\begin{aligned} [0] &= \{x \in \mathbb{Z}: x=3k+0, k \in \mathbb{Z}\} \\ &= \{\dots, -6, -3, 0, 3, 6, \dots\} \end{aligned}$$

$$\begin{aligned} [1] &= \{x \in \mathbb{Z}; x=3k+1, k \in \mathbb{Z}\} \\ &= \{\dots, -5, -2, 1, 4, 7, \dots\} \end{aligned}$$

$$\begin{aligned} [2] &= \{x \in \mathbb{Z}, x=3k+2, k \in \mathbb{Z}\} \\ &= \{\dots, -4, -1, 2, 5, 8, \dots\} \end{aligned}$$

$$\begin{aligned} [3] &= \{x \in \mathbb{Z}, x=3k+3, k \in \mathbb{Z}\} \\ &= \{x \in \mathbb{Z}, x=3(k+1), k \in \mathbb{Z}\} = [0] \end{aligned}$$

Hence, $x \equiv y \pmod{3}$ has three distinct equivalence classes $[0]$, $[1]$ & $[2]$.

Q Define R on $\mathbb{Z}^+$ by $(a, b) R (c, d) \Leftrightarrow a + d = b + c$.

- (1) Show that R is an equivalence
- (2) Find $[(2, 5)]$ i.e. equivalence class of $(2, 5)$
- (3) Find $[(1, 3)]$

Solⁿ (i) $(a, a) R (a, a)$ as $a + a = a + a \quad \forall a \in \mathbb{Z}^+$
 $\therefore$ Reflexive

(ii) $(a, b) R (c, d) \Rightarrow a + d = b + c$
 $c + b = a + d \Rightarrow (c, d) R (a, b)$

$\therefore$ Symmetric

(iii) $(a, b) R (c, d) \text{ \& } (c, d) R (e, f) \Rightarrow a + d = b + c$
 $c + f = d + e$

$$\Rightarrow a + d + c + f = b + c + d + e$$

$$a + f = b + e$$

$$\therefore (a, b) R (e, f)$$

$\Rightarrow R$ is transitive

Hence, R is an equivalence relation.

$$(2) \quad [(2, 5)] = \{ (a, b) : (a, b) R (2, 5), a, b \in \mathbb{Z}^+ \}$$

$$(a, b) R (2, 5) \Rightarrow a + 5 = b + 2;$$

$$\Rightarrow [(2, 5)] = \{ (1, 4), (2, 5), (3, 6), (4, 7), \dots \}$$

$$(3) \quad [(1, 3)] = \{ (a, b) : (a, b) R (1, 3), a, b \in \mathbb{Z}^+ \}$$

$$(1,3) R (a,b) \Leftrightarrow 1+b=3+a$$

$$\text{or } a+3=b+1$$

$$\Rightarrow [1,3] = \{(1,3), (2,4), (3,5), (4,6), \dots\}$$

Q Let $A = \{0, 1, 2, 3, 4\}$ and $R = \{(0,0), (0,4), (1,1), (1,3), (2,2), (3,1), (3,3), (4,0), (4,4)\}$ is an equivalence relation of A . Find distinct equivalence classes of R .

Solⁿ:

$$\left. \begin{aligned} [0] &= \{0, 4\} = [4] \\ [1] &= \{1, 3\} = [3] \\ [2] &= \{2\} \end{aligned} \right\} \text{Distinct}$$

Ex- Let R be a relation congruence modulo 3. Which of the following equivalence classes are equal? $[7], [-4], [-6], [17], [4], [27], [19]$

Solⁿ: $x \equiv y \pmod{3} \Rightarrow x - y = 3k, \quad k \in \mathbb{Z}$

$$7 \equiv x \pmod{3} \Rightarrow 7 = 3k + x \rightarrow x = 1, \quad k = 2$$

$$4 \equiv x \pmod{3} \Rightarrow 4 = 3k + x \Rightarrow x = 1, \quad k = 1$$

$$\Rightarrow [4] = [7]$$

$$19 \equiv x \pmod{3} \Rightarrow 19 = 3k + x \Rightarrow x = 1, \quad k = 6$$

$$\therefore [4] = [7] = [19]$$

$$\begin{aligned} -4 &\equiv x \pmod{3} \Rightarrow -4 = 3k + x \Rightarrow x = 2, k = -2 \\ 17 &\equiv x \pmod{3} \Rightarrow 17 = 3k + x \Rightarrow x = 2, k = 5 \end{aligned}$$

$$\therefore [-4] = [17]$$

$$\begin{aligned} -6 &\equiv 3 \pmod{3} \Rightarrow -6 = 3k + 0 \Rightarrow x = 0, k = -2 \\ 27 &\equiv 3 \pmod{3} \Rightarrow 27 = 3k + 0 \Rightarrow x = 0, k = 9 \end{aligned}$$

$$\therefore [-6] = [27]$$

Partition of a set: A partition of a set A is a set of non-empty subsets of A denoted by $\{A_1, A_2, \dots, A_n\}$ such that

$$(i) \bigcup_{i=1}^n A_i = A \quad (ii) A_i \cap A_j = \emptyset, i \neq j$$

Ex- $A = \{0, 1, 2, 3, 4\}$ then $A_1 = \{0, 1\}$, $A_2 = \{2, 3\}$ and $A_3 = \{4\}$ is a partition of A as $\bigcup_{i=1}^3 A_i = A$ & $A_i \cap A_j = \emptyset$ & $A_i \neq \emptyset$.

Ex- Let $A = \{a, b, c, d, e\}$ and $R = \{(a, b), (a, a), (b, a), (b, b), (c, c), (d, d), (d, e), (e, e), (e, d)\}$ and R be an equivalence relation on A. Determine partition of (i) R (ii) R^{-1}

$$(i) \begin{aligned} [a] &= \{a, b\} = [b] \\ [c] &= \{c\} \end{aligned}$$

$$[d] = \{d, e\} = [e]$$

$$A_1 = \{a, b\}, \quad A_2 = \{c\}, \quad A_3 = \{d, e\}$$

$$\{A_1, A_2, A_3\}$$

$\therefore$ Partition of $R = \{\{a, b\}, \{c\}, \{d, e\}\}$.

ii) Partition of $R =$ Partition of R^{-1}

Ex- Let R be an equivalence relation on set $A = \{1, 2, 3, 4\}$ defined by partitions $P = \{\{1, 4\}, \{2, 3\}\}$. Determine the elements of equivalence relation and also find equivalence classes of R .

Solⁿ: $R = \{(1, 1), (1, 4), (4, 1), (4, 4), (2, 2), (2, 3), (3, 3), (3, 2)\}$

$$[1] = \{1, 4\} = [4]$$

$$[2] = \{2, 3\} = [3]$$

Composition of relations

$$R = \{(a, b), (b, c), (c, a)\} \quad S = \{(b, a), (c, d)\}$$

$$R \circ S =$$

$$S \circ R =$$

It is not commutative but associative.

Closure of Relations

- 1) Graphs
- 2) Matrices
- 3) Composition

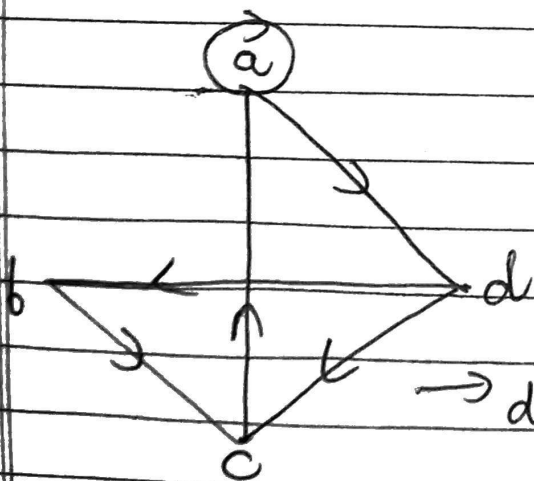
$$G \equiv (V, E)$$

vertex
set

edge
set

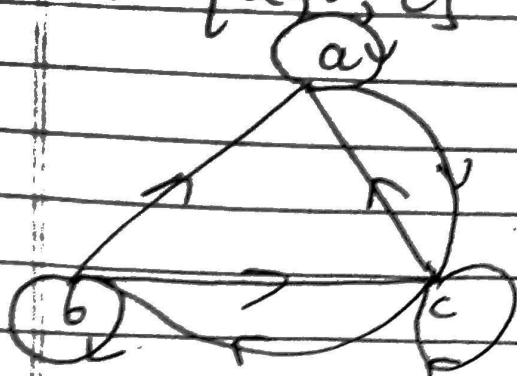
$$V = \{a, b, c, d\}$$

$$R = \{(a, a), (a, d), (b, c), (c, b), (c, a), (d, c)\}$$



→ directed graph / digraph

$$V = \{a, b, c\}$$



$$R = \{(a, a), (a, c), (c, a), (b, c), (c, b), (b, a)\}$$

$$R_{\bar{R}} = \{(a, a), (b, b), (c, c)\}$$

Reflexive closure of R

$$R_{\bar{S}} = R \cup \{(a, b)\}$$

Symmetric closure of R

Closure by composition of relations

$$I = \{(a, a) : a \in A\}$$

$$I_A = \{(a, a) : a \in A\}$$

$$A = \{a, b, c\}$$

$$I_A = \{(a, a), (b, b), (c, c)\}$$

$$R = \{(a, b), (c, a), (b, c), (b, b), (a, c)\}$$

$$R_{\bar{R}} = R \cup I_A$$

$$= \{(a, b), (c, a), (b, c), (b, b), (a, c), (a, a), (c, c)\}$$

Reflexive closure

$$R_{\bar{S}} = R \cup R^{-1}$$

$$= \{(a, b), (c, a), (b, c), (b, b), (a, c), (b, a), (a, c), (c, b), (c, a)\}$$

Reflexive symmetric closure

$$R^n = R^{n-1} \text{ OR } R$$

Transitive closure

$$R^+ = R \cup R^2 \cup R^3 \dots$$

$$R^2 = \{ (a,b), (c,a), (b,c), (b,b), (a,c) \} \cup \{ (a,b), (c,a), (b,c), (b,b), (a,c) \}$$

$$= \{ (a,c), (a,b), (c,b), (c,c), (b,a), (b,c), (b,b), (a,a) \}$$

$$R^3 = R^2 \text{ OR } R$$

$$= \{ (a,c), (a,b), (c,b), (c,c), (b,a), (b,c), (b,b), (a,a) \} \cup \{ (a,b), (c,a), (b,c), (b,b), (a,c) \}$$

$$= \{ (a,a), (a,c), (a,b), (c,c), (c,b), (c,a), (b,b), (b,c), (b,a) \}$$

$$R^+ = R \cup R^2 \cup R^3$$

$$= \{ (a,b), (c,a), (b,c), (b,b), (a,c), (c,b), (c,c), (b,a), (a,a) \}$$

Transitive closure

Q Find reflexive, symmetric and transitive closure of R using composition of relations where R is the relation defined on set A

$$A = \{0, 1, 2, 3\} \text{ by } R = \{ (0,1), (1,2), (2,3) \}$$

$$R^+ = R \cup I_A$$

$$I_A = \{(0,0), (1,1), (2,2), (3,3)\}$$

$$\begin{aligned} R_{\bar{R}} &= \{(0,1), (1,2), (2,3)\} \cup \{(0,0), \\ &\quad (1,1), (2,2), (3,3)\} \\ &= \{(0,1), (1,2), (2,3), (0,0), (1,1), (2,2), \\ &\quad (3,3)\} \end{aligned}$$

$$\begin{aligned} R_5 &= R \cup R^{-1} = \{(0,1), (1,2), (2,3)\} \cup \{ \\ &\quad (1,0), (2,1), (3,2)\} \\ &= \{(0,1), (1,2), (2,3), (1,0), \\ &\quad (2,1), (3,2)\} \end{aligned}$$

$$\begin{aligned} R^2 &= R \circ R = \{(0,1), (1,2), (2,3)\} \circ \{ \\ &\quad (0,1), (1,2), (2,3)\} \\ &= \{(0,2), (1,3)\} \end{aligned}$$

$$\begin{aligned} R^3 &= R^2 \circ R = \{(0,2), (1,3)\} \circ \{(0,1), (1,2), \\ &\quad (2,3)\} \\ &= \{(0,3)\} \end{aligned}$$

$$\begin{aligned} R^4 &= R^3 \circ R = \{(0,3)\} \circ \{(0,1), (1,2), (2,3)\} \\ &= \emptyset \end{aligned}$$

$$R_T = R \cup R^2 \cup R^3 \cup R^4$$

$$= \{(0,1), (1,2), (2,3), (0,2), (1,3), (0,3)\}$$

Matrix representation of R:

$$A = \{a, b, c, d\} \quad R = \{(a,a), (a,d), (d,b), \\ (b,c), (c,c), (d,d)\}$$

$$M_R = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

- If transpose is equal, then relation is symmetric.
- If PD contains all 1, then relation is reflexive.

Reflexive closure

$$M_{\bar{R}} = M_R \cup I_A$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \cup \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$R = \{ (1,1), (1,4), (2,3), (3,3), (4,2), (4,4) \}$$

Symmetric closure

$$M_{\bar{S}} = M_R \cup M_R^T \cong M_R^{-1}$$

$$M_{\bar{S}} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \cup \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$$M_{\bar{R}} = M_R \cup M_{R^2} \cup M_{R^3} \cup M_{R^4}$$

Q Find reflexive, symmetric and transitive closure of R

where $A = \{a, b, c, d\}$

$R = \{(a, b), (b, c), (c, c), (c, a), (c, b)\}$

$$M_{R \cup I_A} = M_R \cup I_A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \cup \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$M_{R \cup I_A} \cup M_{R \cup I_A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \cup \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

18/02/26

Warshall Algorithm : Let $A = \{a_{11}, a_{12}, \dots\}$ be a non-empty set and

R be a relation defined on A .

Let

- 1) $W_0 = M_R$
- 2) Transfer all 1's of W_0 to W_1 (keeping the place intact)
- 3) Consider first row and first column of W_0 . List the locations of $p_1, p_2, \dots, p_r, 1 \leq r \leq n$ in column 1 of W_0 where entry is 1 and locations $q_1, q_2, \dots, q_k, 1 \leq k \leq n$ of row 1 of W_1 where entry is 1.
- 4) Place 1 at the location (p_i, q_j) if 1 is already not there, hence we get W_1
- 5) Repeat steps 2 to 4 with II row and II column to get W_2
- 6) Repeat till all rows and columns are discovered

Ex- Find transitive closure of $R = \{(0,1), (1,3), (3,0)\}$ on $A = \{0,1,2,3\}$.

$\rightarrow W_0 = M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$

$C_1 \rightarrow 4$
 $R_1 \rightarrow 2$
 $(C, R) \equiv (4, 2)$

$W_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

$C_2 \rightarrow 1, 4$
 $R_2 \rightarrow 4$
 $(C, R) \equiv (1, 4), (4, 4)$

$$W_2 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$C_3 \rightarrow 4$$

$$R_3 \rightarrow \phi$$

$$W_3 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$C_4 \rightarrow 1, 2, 3$$

$$R_4 \rightarrow 1, 2, 3, 4$$

$$C_1(1), C_1(2), C_1(3), C_1(4), \\ C_2(1), C_2(2), C_2(3), C_2(4), \\ C_4(1), C_4(2), C_4(3), C_4(4)$$

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

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$$W_0 = M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = W_1$$

2nd col $\rightarrow$ 1
2nd row $\rightarrow$ 3
(1,3)

$$W_2 = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3rd col $\rightarrow$ 1, 2
3rd row $\rightarrow$ 4
(1,4), (2,4)

$$W_3 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = W_4$$

$$R_T = \{ (a,b), (a,c), (a,d), (b,c), (b,d), (c,d) \}$$

Q Let $R = \{ (a,b), (b,c), (c,d), (c,a), (c,b) \}$ defined on set $A = \{ a, b, c \}$.

$$W_0 = M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

1st col $\rightarrow$ 3, 1st row $\rightarrow$ 2 (3,2)

$$W_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

2nd col $\rightarrow$ 1, 3
2nd row $\rightarrow$ 3
(1,3), (3,3)

$$W_2 = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

3rd col. $\rightarrow 1, 2, 3$
3rd row $\rightarrow 1, 2, 3$

$$W_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$C(1,1) \dots C(1,3)$
 $C(2,1) \dots C(2,3)$
 $C(3,1) \dots C(3,3)$

$$R_T = \{ (a,a), (a,b), (a,c), (b,a), (b,b), (b,c), \\ (c,a), (c,b), (c,c) \}$$